$\LaTeX$-Vorlage für Bachelor-, Studien- und Masterarbeiten

Masterarbeit

vorgelegt von

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2025

Bearbeitungszeitraum	8 Wochen
Matrikelnummer, Kurs	XXX, YYY
Ausbildungsbetrieb	ZIEHL-ABEGG SE, Künzelsau
Betreuer der Ausbildungsfirma	XXX

Sperrvermerk

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Künzelsau, 1. Januar 2025

Sven Beller

Erklärung

Ich, Name, versichere hiermit, dass ich meine Hier T3_2000/T3_3100 etc mit dem Thema Hier Thema selbstständig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel benutzt habe.

Ich erkläre zudem, dass ich generative Systeme (Text-, Bild-, Codeerstellung etc.) bei der Erstellung der vorliegenden wissenschaftlichen Arbeit in der nachfolgend beschriebenen Art und dem nachfolgend beschriebenen Umfang verwendet habe.

Ort, Datum

Name

Genutzter Funktionsumfang

Themenerfassung und Strukturierung (Aufgabenstellung/Präzisierung, Forschungsansatz, Gliederung)

Produkt/Tool	Beschreibung / Verwendungsweise
Tool	Ideenfindung und Grobgliederung der Arbeit. Beispiel: Welche Themen existieren über Thema?

Quellenrecherche, -auswahl und -auswertung (durch AI gefundene Quellen; Vorgehen der weiteren Recherche)

Produkt/Tool	Beschreibung / Verwendungsweise
Tool	Vorschläge relevanter Publikationen, anschließende manuelle Validierung. Beispiel: Welche Bücher behandeln das Thema Thema?

Formale Gestaltung, insbesondere Sprache (Prompts/Eingaben; Textgenerierung, Korrektur, Paraphrasieren, Ãœbersetzen, kreatives Schreiben)

Produkt/Tool	Beschreibung / Verwendungsweise
Tool	Grammatik-/Stilkorrektur einzelner Absätze; keine inhaltliche Neuerstellung. Beispiel: Ist der folgende Satz grammatikalisch richtig?

Abstract

This is the abstract in English.

Zusammenfassung

Dies ist die Zusammenfassung auf Deutsch.

Inhaltsverzeichnis

Abbildungsverzeichnis

Tabellenverzeichnis

Nomenklatur

Lateinische Buchstaben

A	m^2	Fläche
$\vec{a}$	$\mathrm{m/s}^2$	Beschleunigung
a	$\mathrm{m/s}$	Schallgeschwindigkeit
c_p	$\mathrm{J/(kg\,K)}$	spezifische isobare Wärmekapazität
c_v	$\mathrm{J/(kg\,K)}$	spezifische isochore Wärmekapazität
E	J	innere Energie
e	$\mathrm{J/kg}$	spezifische innere Energie
g	$\mathrm{m/s}^2$	Erdbeschleunigung
H	J	Enthalpie
h	$\mathrm{J/kg}$	spezifische Enthalpie
k	$\mathrm{J/K}$	Boltzmann-Konstante $\left(k = 1{,}3807 \cdot 10^{-23}\right)$
m	kg	Masse
N_A	mol^{-1}	Avogadro-Konstante $\left(N_\mathrm{A} = 6{,}0021318 \cdot 10^{23}\right)$
$\vec{n}$	-	nach außen gerichteter Normaleneinheitsvektor
n	-	Polytropenexponent
P	$\mathrm{J/s}$	Leistung
p	$\mathrm{N/m}^2$	Druck
Q	J	Wärme
$\dot{Q}$	$\mathrm{J/s}$	Wärmestrom
q	$\mathrm{J/kg}$	bezogene Wärme
$\dot{q}$	$\mathrm{J/(kg\,s)}$	bezogener Wärmestrom
R	$\mathrm{J/(kg\,K)}$	spezielle Gaskonstante
R_m	$\mathrm{J/(mol\,K)}$	universelle Gaskonstante $\left(R_\mathrm{m} = 8{,}31441\right)$
T	K	Temperatur
t	s	Zeit
u, v, w	$\mathrm{m/s}$	Geschwindigkeitskomponenten im kartesischen Koordinatensystem
$\vec{V}$	$\mathrm{m/s}$	Geschwindigkeitsvektor
V	m^3	Volumen
v	$\mathrm{m^3/kg}$	spezifisches Volumen
W	J	Arbeit
x, y, z	m	kartesische Koordinaten
z	m	Höhe

Griechische Buchstaben

η	Pa s	dynamische Viskosität (Scherviskosität)
κ	-	Isentropenexponent
ν	m^2/s	kinematische Viskosität
ν	-	Polytropenverhältnis
ϱ	kg/m^3	Dichte
τ_w	N/m^2	Wandschubspannung
$\vec{\omega}$	s^{-1}	Winkelgeschwindigkeit

Tiefgestellte Indizes

0	Ruhegröße
n	normal
t	tangential

Hochgestellte Indizes

$*$	kritischer Zustand (LAVAL-Zustand)

Dimensionslose Kennzahlen

Ma	:=	$\Vert\vec{V}\Vert/a$	Mach-Zahl
Ma^*	:=	$\Vert\vec{V}\Vert/a^*$	kritische Mach-Zahl
Re	:=	$\Vert\vec{V}\Vert L/\nu$	Reynolds-Zahl

Operatoren, Funktionen und Symbole

$\vert\bullet\vert$	Betrag eines Skalars
$\Vert\bullet\Vert$	Norm (Länge) eines Vektors
d	totales bzw. vollständiges Differential
$\det(\bullet)$	Determinante
D	materielles (substantielles) Differential
$\mathrm{grad}(\bullet)$	Gradient einer tensoriellen Größe
δ	unvollständiges Differential
∂	partielles Differential
∂V	stückweise glatter Rand eines Volumens
∇	Nablaoperator
$\cdot$	Skalarprodukt (inneres Produkt)

$\times$	Kreuzprodukt (äußeres Produkt)

Abkürzungen

EC-Motor	Bürstenloser Gleichstrommotor (Electronically Commutated Motor)
FFT	Schnelle Fourier Transformation (Fast Fourier Transformation)
HSU	Kunstkopf-Messeinheit (Head Shoulder Unit)
RMS	Effektivwert (Root Mean Square)

1 Analyse

Text

1.1 Analyse 1

Quellenangabe [1, S. 2] oder diese Quelle [2, S. 10 ff.]

1.1.1 Freddy

Yes

Fazbear

Text

Mehr Text

Mehr Text

2 Einleitung

2.1 Motivation

Das Abkürzungsverzeichnis in der Nomenklatur wird nur ergänzt, wenn eine Abkürzung auch tatsächlich verwendet wird, z. B. Schnelle Fourier Transformation (Fast Fourier Transformation) (FFT), Bürstenloser Gleichstrommotor (Electronically Commutated Motor) (EC-Motor) oder Kunstkopf-Messeinheit (Head Shoulder Unit) (HSU).

Wer jedoch nur das Akronym verwenden möchte, das dann ebenfalls im Abkürzungsverzeichnis auftaucht, der kann `\acs` verwenden, z. B. RMS.

2.2 Zielsetzung

tbd

2.2.1 Tabelle

Test	Tester
1	2
4	3

Tabelle 2.1: Tabelle als Beispiel

2.2.2 Image

Abb. 2.1: Ein Bild Beispiel

3 Grundlagen

3.1 Grundlagen der Akustik

CTRL + C, dann CTRL + V, fertig.

3.2 Grundlagen der Fourier

$$X(k) = \sum_{n=0}^{N-1} x(n) \cdot e^{-i2\pi nk/N}, \quad k = 0, 1, \ldots, N-1 \tag{3.1}$$

3.3 Verschiedene Algorithmen

Der implementierte BubbleSort-Algorithmus wird in Algorithmus 1 dargestellt. Diese Implementation enthält zwei Optimierungen. Zu Beginn jedes Durchlaufs wird in Zeile 2. die Boolean-Variable `swapped` auf **false** gesetzt. Dies signalisiert dem Algorithmus, dass das Array bereits sortiert ist, solange keine Vertauschungen mehr erfolgen. Eine weitere Modifikation findet sich in der zweiten `for`-Schleife mit Bedingung `n - i - 2`.

Algorithm 1 BubbleSort

Require: Array A mit n Elementen
Ensure: Sortiertes Array A
1: **for** $i = 0$ to $n-1$ **do**
2:     Setze *swapped* auf **false**
3:     **for** $j = 0$ to $n-i-2$ **do**
4:         **if** $A[j] > A[j+1]$ **then**
5:             Vertausche $A[j]$ und $A[j+1]$
6:             Setze *swapped* auf **true**
7:         **end if**
8:     **end for**
9:     **if** nicht *swapped* **then**
10:         **break**
11:     **end if**
12: **end for**

4 Napoleonische Kriege

The Napoleonic Wars (1803-1815) were a global series of conflicts fought by a fluctuating array of European coalitions against the French First Republic (1803-1804) under the First Consul followed by the First French Empire (1804-1815) under the Emperor of the French, Napoleon Bonaparte. The wars originated in political forces arising from the French Revolution (1789-1799) and from the French Revolutionary Wars (1792—1802) and produced a period of French domination over Continental Europe.[19][page needed] The wars are categorised as seven conflicts, five named after the coalitions that fought Napoleon, plus two named for their respective theatres: the War of the Third Coalition, War of the Fourth Coalition, War of the Fifth Coalition, War of the Sixth Coalition, War of the Seventh Coalition, the Peninsular War, and the French invasion of Russia.[20]

The first stage of the war broke out when Britain declared war on France on 18 May 1803, alongside the Third Coalition. In December 1805, Napoleon defeated the allied Russo-Austrian army at Austerlitz, which led to the dissolution of the Holy Roman Empire and thus forced Austria to make peace. Concerned about increasing French power, Prussia led the creation of the Fourth Coalition, which resumed war in October 1806. Napoleon defeated the Prussians at Jena-Auerstedt and the Russians at Friedland, bringing an uneasy peace to the continent. The treaty had failed to end the tension, and war broke out again in 1809, with the Austrian-led Fifth Coalition. At first, the Austrians won a significant victory at Aspern-Essling but were quickly defeated at Wagram.

Hoping to isolate and weaken Britain economically through his Continental System, Napoleon launched an invasion of Portugal, the only remaining British ally in continental Europe. After occupying Lisbon in November 1807, and with the bulk of French troops present in Spain, Napoleon seized the opportunity to turn against his former ally, depose the reigning Spanish royal family, and declare his brother as Joseph I the King of Spain in 1808. The Spanish and Portuguese then revolted with British support, and expelled the French from Iberia in 1814 after six years of fighting.

Concurrently Russia, unwilling to bear the economic consequences of reduced trade, routinely violated the Continental System, prompting Napoleon to launch a massive invasion in 1812. The resulting campaign ended in disaster for France and the near-destruction of Napoleon's Grande Armée.

Encouraged by the defeat, Great Britain, Austria, Prussia, Sweden, and Russia formed the Sixth Coalition and began a campaign against France, decisively defeating Napoleon at Leipzig in October 1813. The allies then invaded France from the east, while the Peninsular War spilled over into southwestern France. Coalition troops captured Paris at the end of March 1814, forced Napoleon to abdicate in April, exiled him to the island of Elba, and restored power to the Bourbons. Napoleon escaped from exile in February 1815 and reassumed control of France for around one hundred days, igniting the eponymous conflict.

The allies formed the Seventh Coalition, which defeated him at Waterloo in June 1815 and exiled him to the island of Saint Helena, where he died six years later in 1821.[21]

The wars had profound consequences on global history, including the spread of nationalism and liberalism, advancements in civil law, the rise of Britain as the world's foremost naval and economic power, the appearance of independence movements in Spanish America and the subsequent decline of the Spanish and Portuguese Empires, the fundamental reorganization of German and Italian territories into larger states, and the introduction of radically new methods of conducting warfare. After the end of the Napoleonic Wars, the Congress of Vienna redrew Europe's borders and brought a relative peace to the continent, lasting until the Revolutions of 1848 and the Crimean War in 1853.

5 Battle of Kircholm

The Battle of Kircholm (Lithuanian: Salaspilio m?šis; Polish: Bitwa pod Kircholmem; Swedish: Slaget vid Kirkholm; 27 September [O.S. 17 September] 1605) was one of the major battles in the Polish-Swedish War of 1600-1611. The battle was decided in 20 minutes by a devastating charge of Polish-Lithuanian cavalry, including the Winged Hussars.[7] The battle ended in a decisive victory of the Polish-Lithuanian forces, and is remembered as one of the greatest triumphs of Commonwealth cavalry.

5.1 Background

On 27 September 1605, the Commonwealth and Swedish forces met near the small town of Kircholm (now Salaspils in Latvia, some 18 kilometres (11 mi) south-east of Riga). The forces of Charles IX of Sweden were numerically superior and were composed of 10,868 men[1] and 11 cannons. The Swedish army included two western commanders, Frederick of Lüneburg and Count Joachim Frederick of Mansfeld,[1] with a few thousand German and Dutch mercenaries and even a few hundred Scots.

The Polish Crown declined to raise funds for defence and send troops, only making promises they never fulfilled.[8] The army, led by the Great Hetman of Lithuania Jan Karol Chodkiewicz, was tired and starving; however, the soldiers admired their leader.[8] He promised to pay army wages from his own fortune, resulting in an influx of recruits from Lithuania. The Polish-Lithuanian Commonwealth army under Chodkiewicz was composed of roughly 1,000 infantry and 2,600 cavalry,[1] but only five cannons. However, the Polish-Lithuanian forces were well-rested, and their cavalry consisted mostly of superbly trained Winged Hussars, or heavy cavalry armed with lances.[citation needed] In contrast, the Swedish cavalry were less-well trained, armed with pistols and carbines, on poorer horses and tired after a long night's march of over 10 km in torrential rain.[3] The Polish-Lithuanian forces had a small number of Lithuanian Tatar and Cossack mercenaries, used mostly for reconnaissance.[citation needed]

5.2 Deployment

Position of both armies prior to Polish-Lithuanian attack

The Swedish forces seem to have been deployed in a checkerboard formation, made up of the infantry regiments formed into seven or eight well-spaced independent blocks, with intersecting fields of fire. The flanks were to be covered by the Swedish and German cavalry, and the cannons were placed in front of the cavalry. Charles' force was formed into four lines on the crest of a ridge. The first line consisted of four infantry battalions, cavalry in the second line, six infantry battalions in the third line, and cavalry in the fourth line.[1] The infantry battalions formed in squares of 30 by 30, with pikemen in the center and shot on the edges, and gaps between the squares allowed passage of their cavalry.[1]

Jan Karol Chodkiewicz deployed his forces in the traditional deep Polish battle formation—the so-called Öld Polish Order"—with the left wing significantly stronger and commanded by Tomasz D?browa, while the right wing was composed of a smaller number of hussars under Jan Piotr Sapieha, and the center included Hetman Chodkiewicz's own company of 300 hussars led by Lt. Wincenty Woyna and a powerful formation of reiters sent by the Duke of Courland Friedrich Kettler.[9] The Polish-Lithuanian infantry, mostly armed in Hungarian hajduk-style, drew up in the center. Some 280 hussars were left as a general reserve under Teodor Lacki.

6 Freddy Fazbear

Five Nights at Freddy's (FNaF) is a video game series and media franchise created by Scott Cawthon that includes video games, novels, graphic novels, and films. The story arcs typically follow a night guard or other character trying to survive from midnight to 6 a.m. for five levels, called "nights", while fending off attacks from homicidal animatronic characters. Each game is set in a different location connected to a fictional pizza restaurant franchise called "Freddy Fazbear's Pizza". The core gameplay mechanics involve using tools effectively and managing limited resources to avoid being caught by the animatronics.

6.1 Bomboclat

Jamaika

6.1.1 Brr brr Patapim

Brainrot

7 Fazit und Ausblick

7.1 Fazit

tbd

7.2 Ausblick

tbd

Literatur

[1] Gh. Reza Sinambari. *Ingenieurakustik*. Springer Vieweg, 2014.

[2] *Akustik – Bestimmung der Schallleistungs- und Schallenergiepegel von Geräuschquellen aus Schalldruckmessungen – Verfahren der Genauigkeitsklasse 1 für reflexionsarme Räume und Halbräume*. Beuth Verlag. DIN EN ISO 3745:2017-10. `https://www.beuth.de`, 2017.

A Anhang A

Weitere Abbildungen